Bivariate Hypothesis Testing

October 10, 2022

As you come in...

Load in the ANES data we used last class (if this code doesn't work, replace the arrow with an equals sign):

```
anes< -read.csv(https://raw.githubusercontent.com/
Brandon-Bolte/PLSC309_F22/main/ANES.csv)
```

Then:

1. Create a boxplot of the feeling thermometer support for the Supreme Court (sctft)
2. Find the mean of this variable
3. Use the t.test() function to find the confidence intervals around this mean
4. Can we be confident that the true population FT score is not 0?
5. Can we be confident that the true population FT score is not 60?

Next Class

In-Class Lab 2

- Read: None
- Due: Checkpoint 2!!!

The Logic of Hypothesis Testing

One-sample test

- We specify what we expect if observed value from sample is no different from a hypothetical population value (null hypothesis)
- We compare the observed value from our sample to the hypothetical value from a population
- The bigger the difference, the more confidently we can reject the null

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Two-sample (bivariate) test

- We specify what we would expect if X and Y are not related in the population (null hypothesis)
- We compare the observe relationship between X and Y from our sample to the expectation under the null
- The bigger the difference between the observed and null relationships, the more confidently we can reject the null

Steps to Testing a Hypothesis

1. State question
2. State the null hypothesis
3. State the alternative hypothesis
4. State the desired confidence level
5. Find critical value
6. Calculate test statistic
7. State conclusion (can we reject the null?)

P-values

P-value: the probability that we would see the observed relationship in our sample data if there were truly no relationship between them in the whole population

- Indicate strength of confidence we can reject the null
- *Lower* p-values suggest *greater* confidence; *higher* values indicate probability is higher that our relationship is due to chance
- Standard cut-off for "statistical significance" is usually $p<0.05$

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BUT:

- Doesn't tell us if relationship is causal
- NOT the same as substantive significance

Bivariate Hypothesis Testing

	Independent Variable		
Dependent Variable		*Categorical*	*Continuous*
	Categorical	Crosstab/Chi-square	Logit/Probit
	Continuous	Difference of Means/T-test; Bivariate Regression	Correlation Coefficient; Bivariate Regression

Cross-Tabulation

Assess the relationship between two nominal variables

We would like to know if the variables are related

- Null hypothesis is that they are *statistically independent*- knowing the value for a unit for one variable does not help us infer anything about that unit's value on the other variable

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The Logic of the Chi-square Test:

- Compare the observed frequencies in a table to expectation under null
- The more these frequencies differ, the more confident we are that variables are related

Example Data

Gender	Democrat	Independent	Republican	Total
Female	573	516	422	1511
Male	386	475	399	1260
Total	959	991	821	2771

Formulating a Hypothesis Test

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Formulating a Hypothesis Test

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Find Critical Value Step 1: Degrees of Freedom

Degrees of freedom represent the maximum number of logically independent values, which are values that have the freedom to vary, in the data sample.

- More degrees of freedom reflect more variation is possible in sample
- In a X^2 test, variation is across cells in crosstab

df = (r-1)(c-1)

r = 2 and c = 3

df = (2-1)(3-1) = 1 x 2 = **2**

Find Critical Value Step 2: Use Table

Upper tail		0.3	0.2	0.1	0.05	0.02	0.01	0.005	0.001
df	1	1.07	1.64	2.71	3.84	5.41	6.63	7.88	10.83
	2	2.41	3.22	4.61	5.99	7.82	9.21	10.60	13.82
	3	3.66	4.64	6.25	7.81	9.84	11.34	12.84	16.27
	4	4.88	5.99	7.78	9.49	11.67	13.28	14.86	18.47
	5	6.06	7.29	9.24	11.07	13.39	15.09	16.75	20.52

Find intersection of confidence level and df for your test

Find Critical Value Step 2: Use Table

Upper tail		0.3	0.2	0.1	0.05	0.02	0.01	0.005	0.001
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	2	2.41	3.22	4.61	5.99	7.82	9.21	10.60	13.82
	3	3.66	4.64	6.25	7.81	9.84	11.34	12.84	16.27
	4	4.88	5.99	7.78	9.49	11.67	13.28	14.86	18.47
	5	6.06	7.29	9.24	11.07	13.39	15.09	16.75	20.52

Find intersection of confidence level and df for your test

- 95% confidence level corresponds to 1-0.95 = 0.05
- 2 degrees of freedom
- *Critical Value* is **5.99**

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4. **I want to test my hypothesis at the 95% confidence level.**
5. **I have 2 degrees of freedom, so my critical value is 5.99**
6. Calculate test statistic
7. Compare test statistic to critical value and state conclusion (can we reject the null?)

Chi-square Test Statistic

$$\chi^2 = \sum \frac{(f_o - f_e)^2}{f_e} \sim \chi^2_{(r-1)(c-1)} \tag{1}$$

- f_o = Frequency observed
- f_e = Frequency expected if no relationship
- r = Number of rows in table
- c = Number of columns in table

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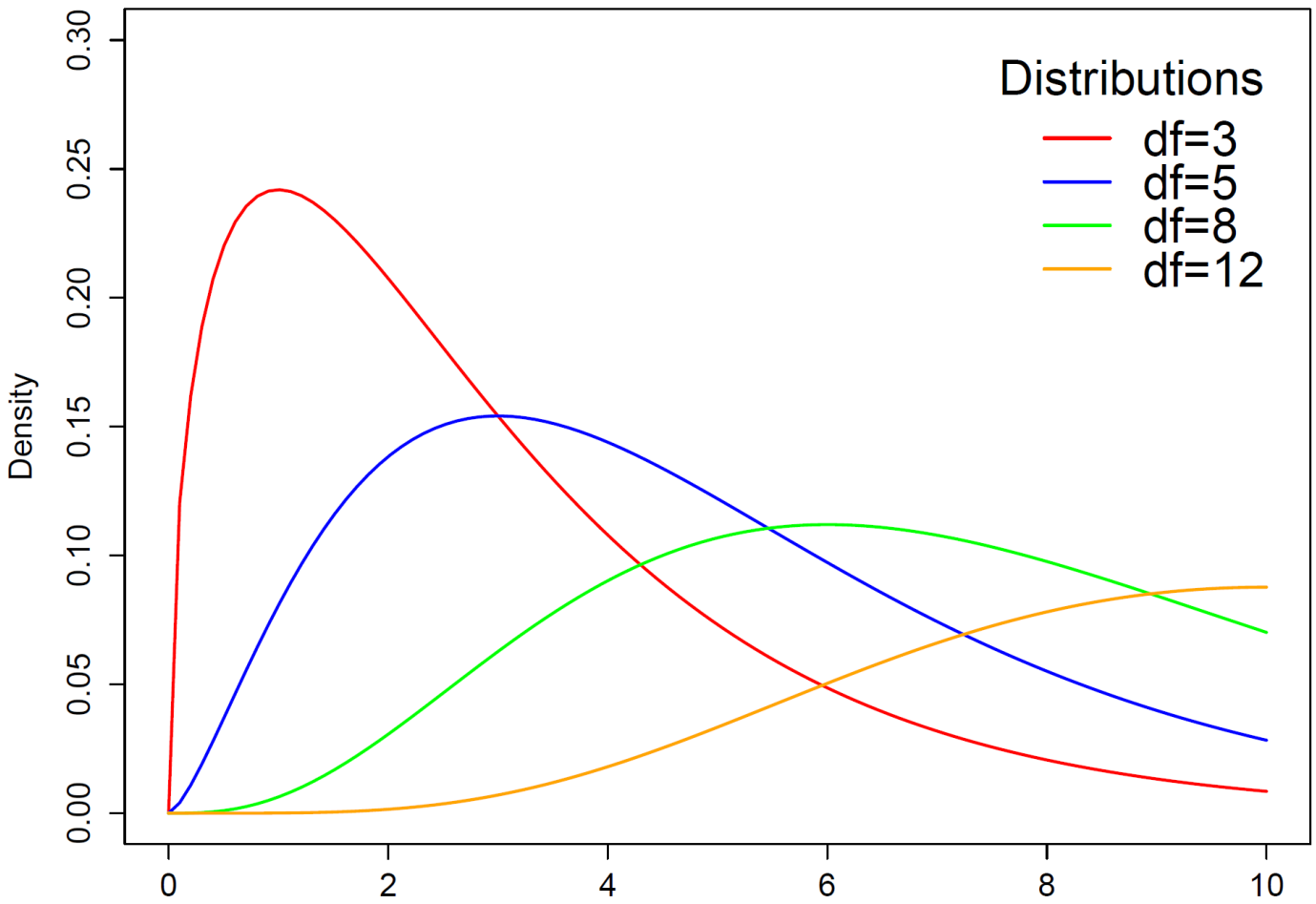
In our case, degrees of freedom is usually $(2-1)(2-1) = 1$

As the differences get bigger, X^2 statistic gets bigger

Chi-square Distribution



Always positive, right-skewed, not symmetric

Shape depends on df- spreads out as df increases

Example Data

Gender	Democrat	Independent	Republican	Total
Female	573	516	422	1511
Male	386	475	399	1260
Total	959	991	821	2771

X^2 Test: Observed Frequency

The number of observations you observed in each cell (the ones that appear in the table)

Gender	Democrat	Independent	Republican	Total
Female	573	516	422	
Male	386	475	399	
Total				

X^2 Test: Expected Frequency

The number of observations you would expect to see in a cell if the two variables are independent (unrelated) from each other (calculate using row and column totals)

Gender	Democrat	Independent	Republican	Total
Female				1511
Male				1260
Total	959	991	821	2771

X^2 Test Step 1: Calculate Expected Frequencies

$$f_e = \frac{\text{Column Total}}{\text{Total Observations in Table}} \times \text{Row Total}$$

Calculate Expected Number of Female Democrats

Gender	Democrat	Independent	Republican	Total
Female	573	516	422	1511
Male	386	475	399	1260
Total	959	991	821	2771

$$\frac{959}{2771} \cdot 1511 = 0.346(1511) = 522.9$$

X^2 Test Step 1: Calculate Expected Frequencies

$$f_e = \frac{\text{Column Total}}{\text{Total Observations in Table}} \times \text{Row Total}$$

Calculate Expected Number of Female Republicans

Gender	Democrat	Independent	Republican	Total
Female	573	516	422	1511
Male	386	475	399	1260
Total	959	991	821	2771

X^2 Test Step 1: Calculate Expected Frequencies

$$f_e = \frac{\text{Column Total}}{\text{Total Observations in Table}} \times \text{Row Total}$$

Calculate Expected Number of Female Republicans

Gender	Democrat	Independent	Republican	Total
Female	573	516	422	1511
Male	386	475	399	1260
Total	959	991	821	2771

$$\frac{821}{2771} \cdot 1511 = 447.7$$

X^2 Test Step 1: Calculate Expected Frequencies

$$f_e = \frac{\text{Column Total}}{\text{Total Observations in Table}} \times \text{Row Total}$$

Calculate Expected Number of Male Independents

Gender	Democrat	Independent	Republican	Total
Female	573	516	422	1511
Male	386	475	399	1260
Total	959	991	821	2771

X^2 Test Step 1: Calculate Expected Frequencies

$$f_e = \frac{\text{Column Total}}{\text{Total Observations in Table}} \times \text{Row Total}$$

Calculate Expected Number of Male Independents

Gender	Democrat	Independent	Republican	Total
Female	573	516	422	1511
Male	386	475	399	1260
Total	959	991	821	2771

$$\frac{991}{2771} \cdot 1260 = 450.6$$

X^2 Test Step 1: Calculate Expected Frequencies

Do this for each cell in the table (expected frequencies are in parentheses here)

Gender	Democrat	Independent	Republican	Total
Female	573	516	422	1511
	(522.9)	(540.4)	(447.7)	
Male	386	475	399	1260
	(436.1)	(450.6)	(373.3)	
Total	959	991	821	2771

X^2 Test Step 2: Calculate Test Statistic

$$
\begin{aligned}
\chi^2 &= \sum \frac{(f_o - f_e)^2}{f_e} \\
&= \frac{(573 - 522.9)^2}{522.9} + \frac{(516 - 540.4)^2}{540.4} + \frac{(422 - 447.7)^2}{447.7} \\
&+ \frac{(386 - 436.1)^2}{436.1} + \frac{(475 - 450.6)^2}{450.6} + \frac{(399 - 373.3)^2}{373.3} \\
&= 4.8 + \ldots + 1.8 \\
&= 16.2
\end{aligned}
$$

Formulating a Hypothesis Test

1. **We want to know if gender is related to party affiliation.**
2. **The null hypothesis is that gender and party affiliation are independent (unrelated).**
3. **The alternative hypothesis is that gender and party affiliation are not independent of each other.**
4. **I want to test my hypothesis at the 95% confidence level.**
5. **I have 2 degrees of freedom, so my critical value is 5.99**
6. **My test statistic is** $X^2 = 16.2$
7. Compare test statistic to critical value and state conclusion (can we reject the null?)

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1. **We want to know if gender is related to party affiliation.**
2. **The null hypothesis is that gender and party affiliation are independent (unrelated).**
3. **The alternative hypothesis is that gender and party affiliation are not independent of each other.**
4. **I want to test my hypothesis at the 95% confidence level.**
5. **I have 2 degrees of freedom, so my critical value is 5.99**
6. **My test statistic is** $X^2 = 16.2$
7. **Since 16.2 is greater than 5.99, we can reject the null hypothesis that gender and party affiliation are statistically independent at the 95% confidence level.**

“Significance”

Statistical Significance: There is a low probability that the relationship we are observing is deviant from the actual relationship in the population

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Theoretical/Political Significance: Important in the real world

X^2 and P-values in R

If we want an exact p-value, we can use R:

- pchisq(**Test Statistic**, **DF**, lower=FALSE)
- pchisq(16.2, 2, lower=FALSE)
- p=0.0003035391

We can also conduct a X^2 test in R:

- chisq.test(**Variable 1**,**Variable 2**,correct=FALSE)

Bivariate Hypothesis Testing

	Independent Variable		
Dependent Variable		*Categorical*	*Continuous*
	Categorical	Crosstab/Chi-square	Logit/Probit
	Continuous	Difference of Means/T-test; Bivariate Regression	Correlation Coefficient; Bivariate Regression

Bivariate t-test: Difference of Means

- One categorical variable and one continuous variable
- Is the mean of the continuous variable in one category the same as the mean in the other category?
  - In other words, are they plausibly drawn from the same underlying distribution in the population?

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- Is the mean of the continuous variable in one category the same as the mean in the other category?
  - In other words, are they plausibly drawn from the same underlying distribution in the population?

Null hypothesis: the sample means are drawn from the same underlying distribution

Alternative hypothesis: the sample means are not drawn from the same underlying distribution (e.g., the means are very different from each other)

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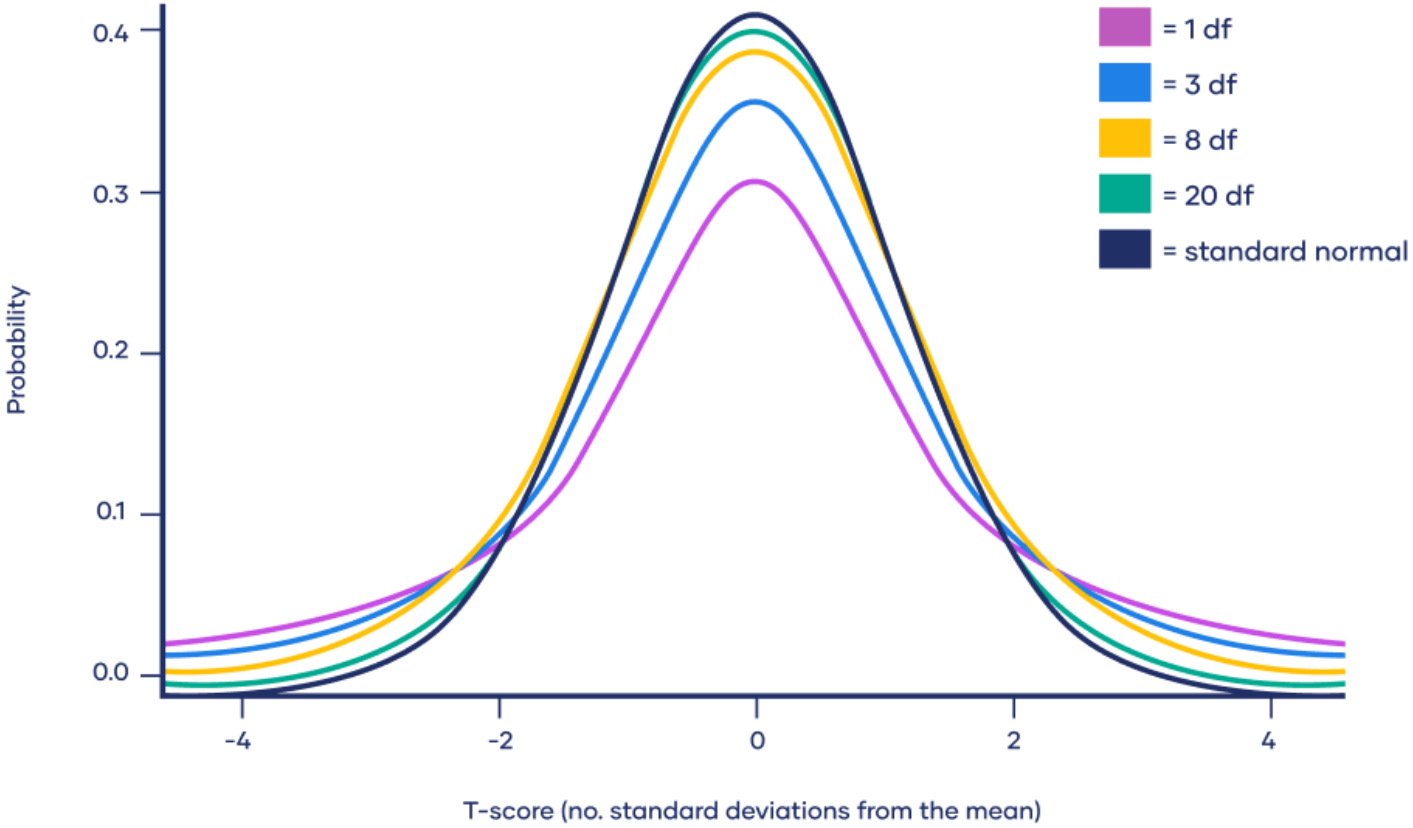
T-test Statistic

$$t = \frac{\bar{Y}_1 - \bar{Y}_2}{se(\bar{Y}_1 - \bar{Y}_2)} \sim t_{n_1+n_2-2} \tag{2}$$

- $\bar{Y}_i$ = the mean of each group i
- $se(\bar{Y}_1 - \bar{Y}_2)$ = standard error of difference of means
- n_i = size of each group i

The t-distribution



One-sample t-test: $df = n - 1$

Two-sample t-test: $df = n_1 + n_2 - 2$